

hw5

Q3

The ratdrink data consist of five weekly measurements of body weight for 27 rats. The first 10 rats are on a control treatment while 7 rats have thyroxine added to their drinking water. Ten rats have thiouracil added to their water.

a

Plot the data showing how weight increases with age on a single panel, taking care to distinguish the three treatment groups. Now create a three-panel plot, one for each group. Discuss what can be seen.

```
library(faraway)
```

Warning: package 'faraway' was built under R version 4.3.3

```
library(dplyr)
```

Warning: package 'dplyr' was built under R version 4.3.1

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
data(ratdrink)
glimpse(ratdrink)
```

Rows: 135

Columns: 4

```
$ wt      <dbl> 57, 86, 114, 139, 172, 60, 93, 123, 146, 177, 52, 77, 111, 144~
$ weeks   <int> 0, 1, 2, 3, 4, 0, 1, 2, 3, 4, 0, 1, 2, 3, 4, 0, 1, 2, 3, 4, 0,~
$ subject <fct> 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 5,~
$ treat   <fct> control, control, control, control, control, control, control, control,~
```

```
summary(ratdrink)
```

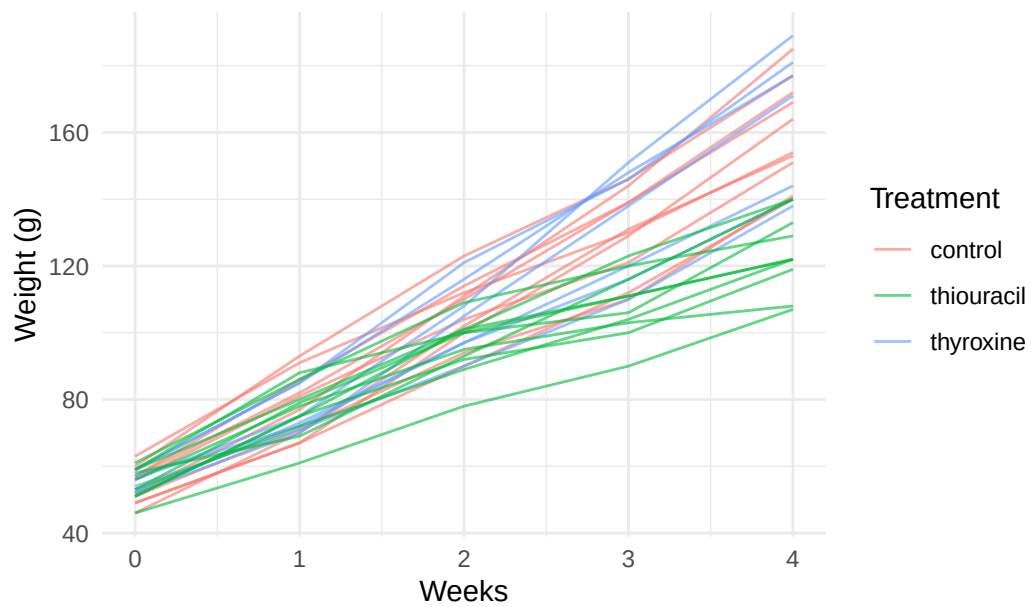
	wt		weeks		subject		treat
Min.	: 46.0	Min.	:0	1	: 5	control	:50
1st Qu.:	71.0	1st Qu.:	1	2	: 5	thiouracil:	50
Median	:100.0	Median	:2	3	: 5	thyroxine	:35
Mean	:100.8	Mean	:2	4	: 5		
3rd Qu.:	122.5	3rd Qu.:	3	5	: 5		
Max.	:189.0	Max.	:4	6	: 5		
					(Other):	105	

```
library(ggplot2)
```

Warning: package 'ggplot2' was built under R version 4.3.1

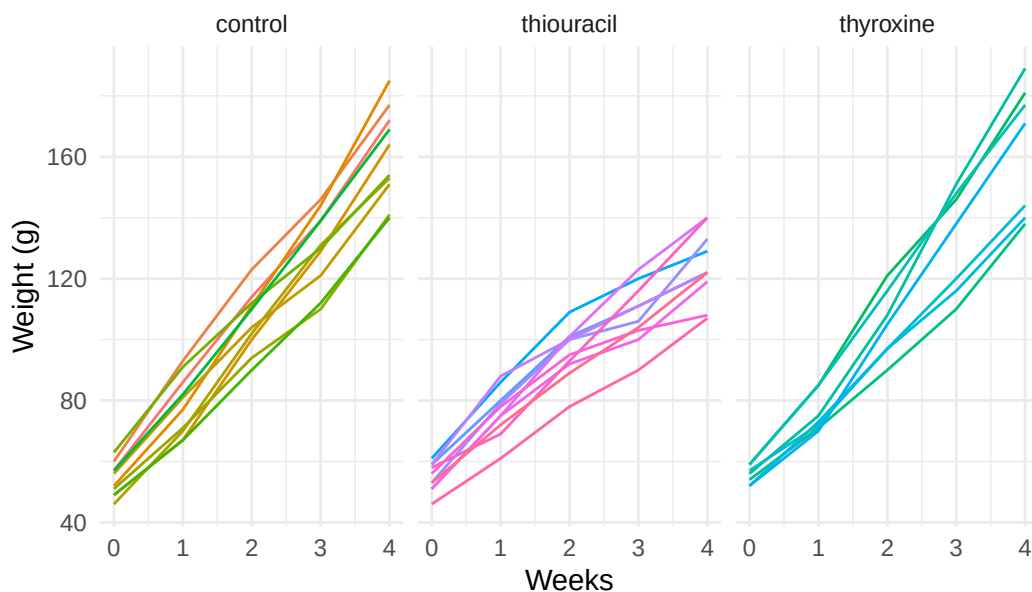
```
# One panel plot
ggplot(ratdrink, aes(x = weeks, y = wt, group = subject, color = treat)) +
  geom_line(alpha = 0.6) +
  labs(title = "Rat Weight Over Time by Treatment",
       x = "Weeks", y = "Weight (g)", color = "Treatment") +
  theme_minimal()
```

Rat Weight Over Time by Treatment



```
# Three-panel faceted plot by treatment
ggplot(ratdrink, aes(x = weeks, y = wt, group = subject)) +
  geom_line(aes(color = subject), show.legend = FALSE) +
  facet_wrap(~ treat) +
  labs(title = "Weight Trajectories by Treatment Group",
       x = "Weeks", y = "Weight (g)") +
  theme_minimal()
```

Weight Trajectories by Treatment Group



From the single panel plot we can see that all groups tend to show increasing weight over time. The second (three-panel) plot shows the weight of rats over time, with different treatments separated via the panels. The three-panel plot allows for a clearer comparison of how each treatment group (control, thyroxine, thiouracil) affects weight gain over the weeks. The control group shows a steady increase in weight, while the thyroxine group appears to gain weight slightly more rapidly. The thiouracil group shows a slower increase, indicating a potential suppression of weight gain compared to the other groups.

b

Fit a linear longitudinal model that allows for a random slope and intercept for each rat. Each group should have a different mean line. Give interpretation for the following estimates:

- The fixed effect intercept term.
- The interaction between thiouracil and week.
- The intercept random effect SD.

```
library(lme4)
```

Warning: package 'lme4' was built under R version 4.3.1

Loading required package: Matrix

Warning: package 'Matrix' was built under R version 4.3.1

```
model_b <- lmer(wt ~ weeks * treat + (weeks | subject), data = ratdrink)
summary(model_b)
```

Linear mixed model fit by REML ['lmerMod']
Formula: wt ~ weeks * treat + (weeks | subject)
Data: ratdrink

REML criterion at convergence: 878.7

Scaled residuals:

	Min	1Q	Median	3Q	Max
	-1.83136	-0.54991	0.04003	0.58230	2.03660

Random effects:

Groups	Name	Variance	Std.Dev.	Corr
subject	(Intercept)	32.49	5.700	
	weeks	14.14	3.760	-0.13
Residual		18.90	4.348	

Number of obs: 135, groups: subject, 27

Fixed effects:

	Estimate	Std. Error	t value
(Intercept)	52.8800	2.0937	25.256
weeks	26.4800	1.2661	20.915
treatthiouracil	4.7800	2.9610	1.614
treatthyroxine	-0.7943	3.2628	-0.243
weeks:treatthiouracil	-9.3700	1.7905	-5.233
weeks:treatthyroxine	0.6629	1.9730	0.336

Correlation of Fixed Effects:

	(Intr)	weeks	trtthr	trtthy	wks:trtthr
weeks		-0.250			
treatthircl	-0.707	0.177			
treatthyrxn	-0.642	0.160	0.454		
wks:trtthrc	0.177	-0.707	-0.250	-0.113	
wks:trtthyr	0.160	-0.642	-0.113	-0.250	0.454

Interpretation of the estimates:

- i. The fixed effect intercept term represents the average weight of a rat at week 0 (the

baseline) across all treatment groups. It serves as the starting point for the weight trajectory of the rats before any treatment effects are applied. At week 0, rats in the control group weigh an average of 52.88 grams.

- ii. The interaction between thiouracil and week indicates how the effect of time (weeks) on weight differs for rats receiving thiouracil compared to the control group. A negative coefficient suggests that as weeks progress, the weight gain in the thiouracil group is less than that in the control group, indicating a potential suppression of weight gain due to thiouracil treatment. Rats in the thiouracil group gain weight at a slower rate than control rats by about 9.37 grams less per week on average, (after adjusting for random individual variation and initial group differences). Even though the thiouracil group has a positive main effect (+4.78), this interaction shows that their growth rate over time is significantly lower than control.
- iii. The intercept random effect SD represents the variability in the initial weights of the rats across different subjects. A higher value indicates that there is considerable individual variation in starting weights among the rats, suggesting that some rats start at a significantly different weight than others, regardless of treatment. In this case, individual rats vary around the group-level intercept (52.88g) with a standard deviation of about 5.7 grams. That is, some rats start heavier or lighter than average, even within the same treatment group.

c

Check whether there is a significant treatment effect.

```
# Full model (same as model_b)
model_full <- lmer(wt ~ weeks * treat + (weeks | subject), data = ratdrink)

# Reduced model (no treatment terms)
model_reduced <- lmer(wt ~ weeks + (weeks | subject), data = ratdrink)

# Likelihood ratio test
anova(model_reduced, model_full)
```

refitting model(s) with ML (instead of REML)

Data: ratdrink

Models:

```

model_reduced: wt ~ weeks + (weeks | subject)
model_full: wt ~ weeks * treat + (weeks | subject)
      npar    AIC    BIC  logLik deviance  Chisq Df Pr(>Chisq)
model_reduced      6 932.85 950.28 -460.42   920.85
model_full        10 915.71 944.76 -447.86   895.71 25.136  4 4.723e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

The likelihood ratio test comparing the reduced model (only time effect) to the full model (which includes treatment and treatment-by-time interactions) yields a Chi-squared statistic of 25.14 on 4 degrees of freedom, with a p-value of 4.723e-05. This provides strong evidence that treatment has a significant effect on weight and including treatment in the model improves the model fit significantly.

d

Construct diagnostic plots showing the residuals against the fitted values and a QQ plot of the residuals. Interpret.

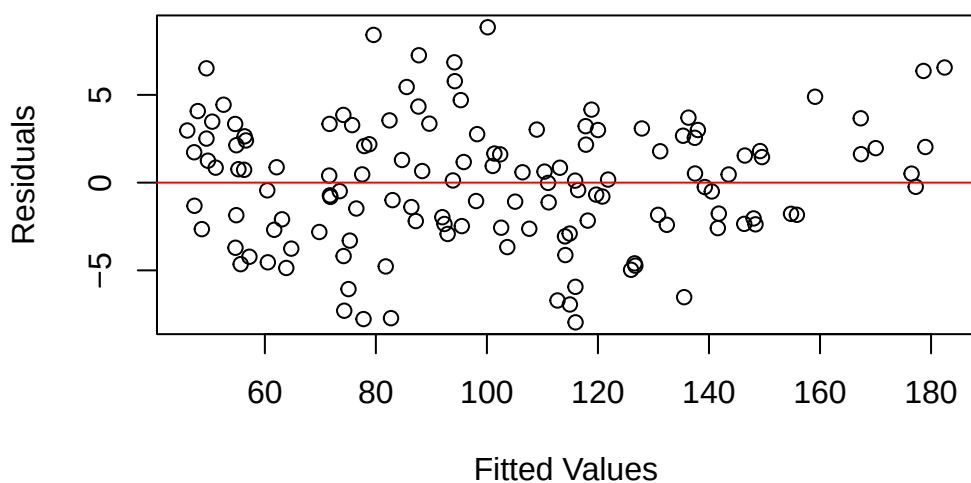
```

resid <- residuals(model_full)
fitted_vals <- fitted(model_full)

# Residuals vs Fitted Plot
plot(fitted_vals, resid,
     xlab = "Fitted Values", ylab = "Residuals",
     main = "Residuals vs Fitted")
abline(h = 0, col = "red")

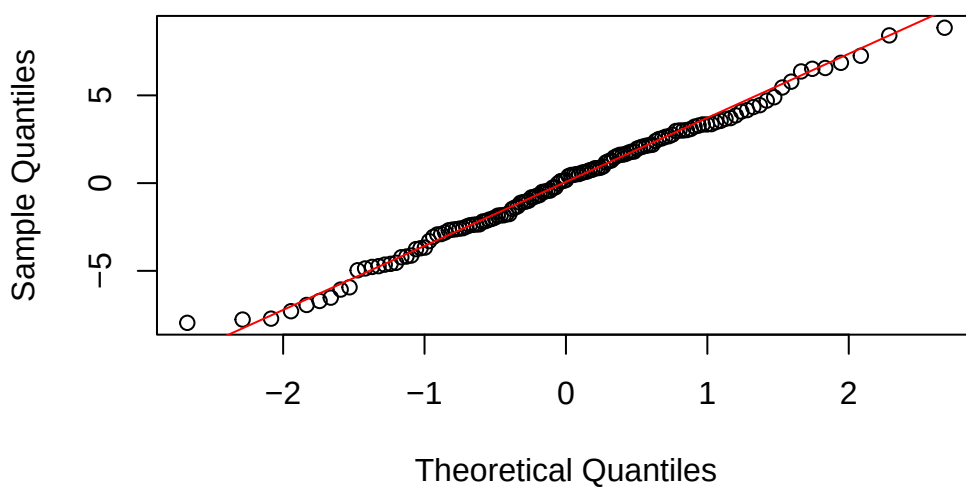
```

Residuals vs Fitted



```
# QQ Plot  
qqnorm(resid, main = "Normal Q-Q Plot of Residuals")  
qqline(resid, col = "red")
```

Normal Q-Q Plot of Residuals



The residuals vs. fitted values plot shows no clear pattern, indicating that the model assumptions of homoscedasticity (constant variance) and linearity are reasonably met. The residuals are randomly scattered around zero, suggesting that the model fits the data well without systematic errors. The QQ plot shows that the residuals mostly follow a straight line, indicating that the assumption of normality of residuals is also reasonably satisfied, though there are very minor deviations at the tails.

e

Construct confidence intervals for the parameters of the model. Which random effect terms may not be significant? Is the thyroxine group significantly different from the control group?

```
ci <- confint(model_full, method = "profile")
```

Computing profile confidence intervals ...

```
rownames(ci)[1:4] <- c("SD: Random Intercept",
                      "Correlation: Intercept/Slope",
                      "SD: Random Slope (weeks)",
                      "Residual SD")
ci
```

	2.5 %	97.5 %
SD: Random Intercept	3.4506344	7.6654748
Correlation: Intercept/Slope	-0.5261017	0.3794396
SD: Random Slope (weeks)	2.6064086	4.8687660
Residual SD	3.7555591	5.1142342
(Intercept)	48.8692859	56.8907140
weeks	24.0547425	28.9052574
treatathiouracil	-0.8920062	10.4520061
treatthyroxine	-7.0445322	5.4559606
weeks:treatathiouracil	-12.7998320	-5.9401680
weeks:treatthyroxine	-3.1166337	4.4423479

Random-intercept SD: The entire CI lies above 0 -> significantly positive.

Correlation (Intercept, slope) : This CI crosses 0 -> the correlation is not significantly different from zero.

Random-slope SD : The entire CI lies above 0 -> significantly positive.

Residual SD : Obviously > 0; not tested for significance in the same way.

Therefore, the only random-effect parameter whose interval includes zero is the intercept-slope

correlation (.sig02). Both variance components (random-intercept SD and random-slope SD) are significantly above zero.

Is the thyroxine group significantly different from the control group?

Intercept (baseline) weight difference (treatthyroxine): 95% CI = [-7.0445, 5.45596] includes 0, so not significantly different from control at week 0.

Slope difference (growth rate) (weeks:treatthyroxine): 95% CI = [-3.1166, 4.44235] includes 0, so not significantly different from control in rate of weight gain.

Q4

The ohio data concern 536 children from Steubenville, Ohio and were taken as part of a study on the effects of air pollution. Children were in the study for 4 years from ages 7 to 10. The response was whether they wheezed or not. The variables are:

resp: an indicator of wheeze status (1 = yes, 0 = no)

id: an identifier for the child

age: 7 yrs = -2, 8 yrs = -1, 9 yrs = 0, 10 yrs = 1

smoke: an indicator of maternal smoking at the first year of the study (1 = smoker, 0 = nonsmoker)

a

Do any of the mothers in the study change their smoking status during the period of observation?

```
data(ohio)
glimpse(ohio)
```

```
Rows: 2,148
```

```
Columns: 4
```

```
$ resp <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
$ id   <int> 0, 0, 0, 0, 1, 1, 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 4, 4, 4, 4, 5, 5~
$ age  <int> -2, -1, 0, 1, -2, -1, 0, 1, -2, -1, 0, 1, -2, -1, 0, 1, -2, -1, ~
$ smoke <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
```

```
summary(ohio)
```

resp	id	age	smoke
Min. :0.0000	Min. : 0	Min. :-2.00	Min. :0.0000
1st Qu.:0.0000	1st Qu.:134	1st Qu.: -1.25	1st Qu.:0.0000
Median :0.0000	Median :268	Median :-0.50	Median :0.0000
Mean :0.1518	Mean :268	Mean :-0.50	Mean :0.3482
3rd Qu.:0.0000	3rd Qu.:402	3rd Qu.: 0.25	3rd Qu.:1.0000
Max. :1.0000	Max. :536	Max. : 1.00	Max. :1.0000

```
# Check if smoke changes within each child
smoke_varies <- ohio %>%
  group_by(id) %>%
  summarize(unique_smoke = n_distinct(smoke)) %>%
  filter(unique_smoke > 1)

# how many such children exist
nrow(smoke_varies)
```

```
[1] 0
```

The data show that no child had a change in maternal smoking status during the 4-year observation period. This confirms that maternal smoking is a fixed variable, measured at baseline and held constant over time in this study.

b

Construct a table that shows proportion of children who wheeze for 0, 1, 2, 3 or 4 years broken down by maternal smoking status.

```
library(tidyr)
```

```
Warning: package 'tidyr' was built under R version 4.3.1
```

```
Attaching package: 'tidyr'
```

```
The following objects are masked from 'package:Matrix':
```

```
expand, pack, unpack
```

```

# Compute total years of wheezing for each child by summing the resp variable
children_wheeze <- ohio %>%
  group_by(id, smoke) %>%
  summarize(
    total_wheeze = sum(resp),
    .groups = "drop"
  )

# Count how many children (within each smoke status) have each total_wheeze
wheeze_counts <- children_wheeze %>%
  group_by(smoke, total_wheeze) %>%
  summarize(
    n_children = n(),
    .groups = "drop"
  )

# Convert counts to proportions within each smoke group
wheeze_props <- wheeze_counts %>%
  group_by(smoke) %>%
  mutate(
    prop = n_children / sum(n_children)
  ) %>%
  ungroup()

# Wide Form Table
wheeze_table_wide <- wheeze_props %>%
  select(smoke, total_wheeze, prop) %>%
  pivot_wider(
    names_from = smoke,
    values_from = prop,
    names_prefix = "smoke_"
  )

wheeze_props

```

```

# A tibble: 10 x 4
  smoke total_wheeze n_children prop
  <int>         <int>     <int> <dbl>
1     0             0       237 0.677
2     0             1        65 0.186
3     0             2        25 0.0714
4     0             3        12 0.0343

```

5	0	4	11	0.0314
6	1	0	118	0.631
7	1	1	32	0.171
8	1	2	19	0.102
9	1	3	11	0.0588
10	1	4	7	0.0374

```
wheeze_table_wide
```

```
# A tibble: 5 x 3
  total_wheeze smoke_0 smoke_1
      <int>    <dbl>   <dbl>
1         0  0.677  0.631
2         1  0.186  0.171
3         2  0.0714 0.102
4         3  0.0343 0.0588
5         4  0.0314 0.0374
```

c

Make plot which shows how the proportion of children wheezing changes by age with a separate line for smoking and nonsmoking mothers.

```
# Convert coded age to actual age
ohio <- ohio %>%
  mutate(real_age = age + 9)

# Compute proportion wheezing at each age by smoking group
wheeze_by_age <- ohio %>%
  group_by(real_age, smoke) %>%
  summarize(
    prop_wheeze = mean(resp),
    .groups = "drop"
  )

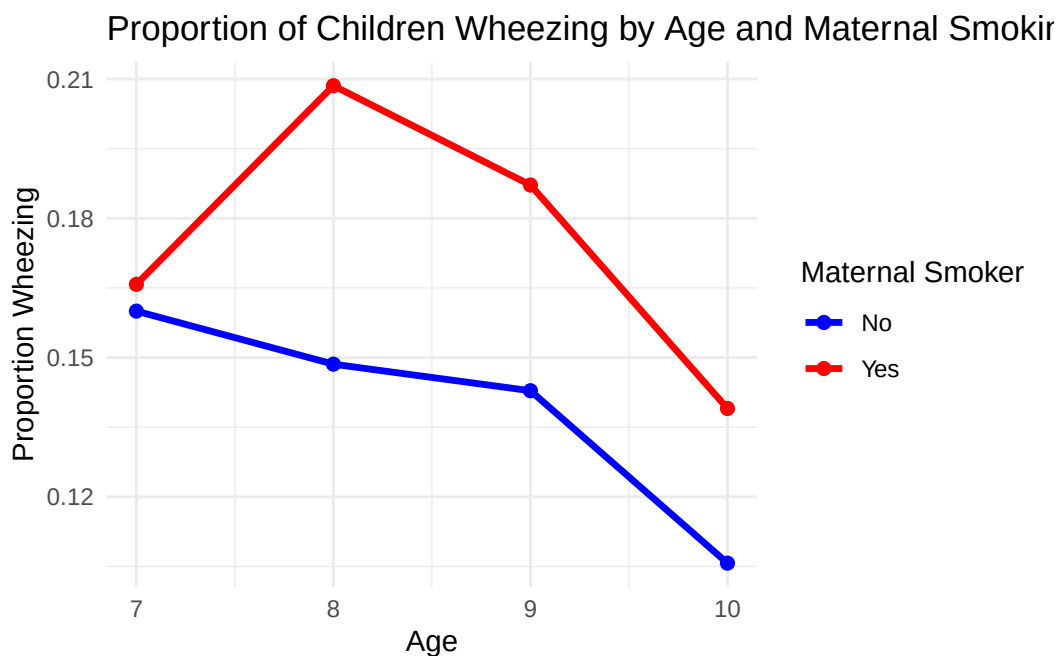
# Plot
ggplot(wheeze_by_age, aes(x = real_age, y = prop_wheeze, color =
  factor(smoke))) +
  geom_line(size = 1.2) +
  geom_point(size = 2) +
  labs(
```

```

title = "Proportion of Children Wheezing by Age and Maternal Smoking",
x = "Age",
y = "Proportion Wheezing",
color = "Maternal Smoker"
) +
scale_x_continuous(breaks = 7:10) +
scale_color_manual(values = c("blue", "red"), labels = c("No", "Yes")) +
theme_minimal()

```

Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.



d

Group the data by child to count the total (out of four) years of wheezing.
Fit a binomial GLM to this response to check for a maternal smoking effect.
Does this prove there is a smoking effect or could there be another plausible explanation?

```

# Count wheezing years per child
wheeze_summary <- ohio %>%
  group_by(id) %>%
  summarize(
    wheeze_years = sum(resp),
    no_wheeze_years = 4 - wheeze_years,
    smoke = first(smoke),
    .groups = "drop"
  )

# Fit binomial GLM
model_binom <- glm(
  cbind(wheeze_years, no_wheeze_years) ~ smoke,
  family = binomial,
  data = wheeze_summary
)

summary(model_binom)

```

Call:

```
glm(formula = cbind(wheeze_years, no_wheeze_years) ~ smoke, family = binomial,
    data = wheeze_summary)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.82124	0.07719	-23.595	<2e-16 ***
smoke	0.27156	0.12334	2.202	0.0277 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 1045.3 on 536 degrees of freedom
 Residual deviance: 1040.5 on 535 degrees of freedom
 AIC: 1337.9

Number of Fisher Scoring iterations: 4

Smoke coefficient: $\log\text{-odds} = 0.272 \rightarrow \text{odds} = \exp(0.272) = 1.313$

Children of smoking mothers have about 1.31 times higher odds of wheezing (in a given year)

compared to children of nonsmoking mothers. Because $p = 0.0277 (< 0.05)$, this association is statistically significant at the 0.05 threshold level.

Does this prove there is a smoking effect or could there be another plausible explanation?

Not by itself. Although the smoking coefficient is significant, this simple binomial GLM cannot by itself prove a causal smoking effect. By collapsing each child's four yearly observations into a single count (ignoring the study's repeated measures structure), it ignores how wheeze risk changes with age and treats each year as independent, but it's likely a child who wheezes once is likelier to wheeze again. It also doesn't adjust for other possibly significant factors (such as socioeconomic status or genetic predisposition) and assumes no extra variability beyond the binomial model. Because of these limitations, a model that accounts for age effects and within-child correlation—such as a GLMM or GEE would be more appropriate for drawing causal conclusions.

e

Fit a model for each individual response using a GLMM fit using penalized quasi-likelihood. Describe the effects of age and maternal smoking. How do the odds of wheezing change numerically over time?

```
library(MASS)
```

Attaching package: 'MASS'

The following object is masked from 'package:dplyr':

```
select
```

```
# Fit the GLMM using PQL
model_pql <- glmmPQL(
  fixed = resp ~ age + smoke,
  random = ~ 1 | id,
  family = binomial,
  data = ohio,
  verbose = FALSE
)

summary(model_pql)
```


Linear mixed-effects model fit by maximum likelihood

Data: ohio

AIC BIC logLik

NA NA NA

Random effects:

Formula: ~1 | id

(Intercept) Residual

StdDev: 2.057175 0.6355563

Variance function:

Structure: fixed weights

Formula: ~invwt

Fixed effects: resp ~ age + smoke

	Value	Std.Error	DF	t-value	p-value
(Intercept)	-2.7658365	0.14218299	1610	-19.452654	0.0000
age	-0.1815756	0.04365164	1610	-4.159652	0.0000
smoke	0.3251839	0.23131699	535	1.405793	0.1604

Correlation:

(Intr) age

age 0.197

smoke -0.591 -0.003

Standardized Within-Group Residuals:

Min	Q1	Med	Q3	Max
-2.6145143	-0.2829352	-0.2583719	-0.2154580	3.4443795

Number of Observations: 2148

Number of Groups: 537

Intercept (log-odds = -2.77): This is the log-odds of wheezing for a child at age 7 with a nonsmoking mother.

Odds of wheezing at that baseline: $\exp(-2.77) = 0.062$, or about 6.2% chance of wheezing at age 7 for children of nonsmoking mothers.

Age (Estimate = -0.18, $p < 0.0001$): or each additional year of age, the log-odds of wheezing decrease by 0.18.

Odds ratio for age: $\exp(-0.18) = 0.835$, meaning the odds of wheezing decrease by about 16.5% for each additional year of age (each year older). This association is also statistically significant due to the extremely low p-value.

Maternal Smoking (Estimate = +0.33, $p = 0.160$): The log-odds of wheezing for children of smoking mothers is 0.33 higher than for children of nonsmoking mothers.

Odds ratio for smoking: $\exp(0.33) = 1.39$, meaning children of smoking mothers have about

39% higher odds of wheezing compared to children of nonsmoking mothers, but this effect is not statistically significant ($p = 0.1604$).

How do the odds of wheezing change numerically over time?

The odds of wheezing decrease by about 16.5% for each additional year of age, indicating that as children get older, they are less likely to wheeze, regardless of maternal smoking status. The effect of maternal smoking is positive but not statistically significant, suggesting that while there may be a trend towards higher wheezing odds in children of smoking mothers, this effect is not strong enough to draw causal conclusions.

f

Now fit the same model but using adaptive Gaussian-Hermit quadrature. Compare to the previous model fit.

```
library(lme4)

# Fit GLMM using AGHQ
model_agq <- glmer(
  resp ~ age + smoke + (1 | id),
  data   = ohio,
  family = binomial,
  nAGQ   = 10
)

summary(model_agq)
```

```
Generalized linear mixed model fit by maximum likelihood (Adaptive
Gauss-Hermite Quadrature, nAGQ = 10) [glmerMod]
Family: binomial ( logit )
Formula: resp ~ age + smoke + (1 | id)
Data: ohio
```

AIC	BIC	logLik	deviance	df.resid
1603.6	1626.3	-797.8	1595.6	2144

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.3682	-0.2024	-0.1785	-0.1498	2.5106

Random effects:

Groups	Name	Variance	Std.Dev.
--------	------	----------	----------

```

id      (Intercept) 4.614    2.148
Number of obs: 2148, groups: id, 537

Fixed effects:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -3.08959    0.21556 -14.333  < 2e-16 ***
age          -0.17533    0.06762  -2.593  0.00952 **
smoke         0.39799    0.27165   1.465  0.14291
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:
      (Intr) age
age    0.246
smoke -0.497 -0.008

```

The GLMM fitted using adaptive Gaussian-Hermite quadrature (AGHQ) produces results that are broadly consistent with the earlier model fitted using penalized quasi-likelihood (PQL). Both models suggest that age has a significant negative effect on wheezing odds, while maternal smoking shows a positive but non-significant effect. One key difference is that the AGHQ model is based on full likelihood estimation and provides more accurate standard errors and valid model comparison criteria such as AIC and BIC. In contrast, the PQL model approximates the likelihood and is known to underestimate standard errors, making it less reliable for inference.

i

Fit the model using GEE. Use an autoregressive rather than exchangeable error structure. Compare the results to the previous model fits. In your model, what indicates that a child who already wheezes is likely to continue to wheeze?

```
library(geepack)
```

```
Warning: package 'geepack' was built under R version 4.3.3
```

```
Attaching package: 'geepack'
```

```
The following object is masked _by_ '.GlobalEnv':
```

```
ohio
```

The following object is masked from 'package:faraway':

ohio

```
# Fit GEE
model_gee_ar1 <- geeglm(
  formula = resp ~ age + smoke,
  data = ohio,
  id = id,
  family = binomial,
  corstr = "ar1"
)

summary(model_gee_ar1)
```

Call:

```
geeglm(formula = resp ~ age + smoke, family = binomial, data = ohio,
       id = id, corstr = "ar1")
```

Coefficients:

	Estimate	Std.err	Wald	Pr(> W)
(Intercept)	-1.90218	0.11525	272.409	<2e-16 ***
age	-0.11489	0.04539	6.407	0.0114 *
smoke	0.23448	0.18119	1.675	0.1956

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation structure = ar1

Estimated Scale Parameters:

	Estimate	Std.err
(Intercept)	1.021	0.1232

Link = identity

Estimated Correlation Parameters:

	Estimate	Std.err
alpha	0.491	0.06733

Number of clusters: 537 Maximum cluster size: 4

The GEE model with an AR(1) correlation structure gives results that are largely consistent with the earlier GLMMs. The age effect remains statistically significant, with an estimated

coefficient of -0.115 (and much less extreme p -value of $p = 0.011$), indicating that each additional year of age is associated with approximately a 10.9% reduction in the odds of wheezing ($OR = 0.89$). The effect of maternal smoking remains positive (estimate = 0.234) but not statistically significant ($p = 0.196$), in line with previous models.

In your model, what indicates that a child who already wheezes is likely to continue to wheeze?

AR(1) correlation ($\alpha = 0.491$): An autoregressive (AR1) correlation structure assumes that repeated measures taken closer in time are more strongly correlated. The estimated AR(1) correlation coefficient, $\alpha = 0.491$, suggests a moderate positive correlation in wheezing responses across years within the same child. This means that if a child wheezes in one year, they are substantially more likely to wheeze in the following year. This autoregressive correlation captures the persistence of wheezing over time and supports the idea that wheezing is not purely random from year to year, but rather tends to reoccur within individuals.

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What is your overall conclusion regarding the effect of age and maternal smoking? Can we trust the GLM result or are the GLMM models preferable?

The overall conclusion regarding the effect of age and maternal smoking:

- Age: There is a consistent and statistically significant negative effect of age on wheezing across all models. As children get older, the odds of wheezing decrease, suggesting that wheezing is more common in younger children and tends to resolve as they grow older.

- Maternal Smoking: The effect of maternal smoking on wheezing is positive but not statistically significant in any of the models. This suggests that while there may be a trend towards higher wheezing odds in children of smoking mothers, the evidence is not strong enough to draw definitive conclusions about a causal relationship.

- Model Preference: The GLMM models are generally preferable to the simple binomial GLM because they account for the repeated measures structure of the data, allowing for individual variability in wheezing responses. The GEE model with an AR(1) structure also provides a robust alternative that captures the correlation of repeated measures within individuals. The GLMMs (especially with adaptive Gauss–Hermite quadrature) and GEE models provide more reliable standard errors and better reflect the underlying data structure, making them more trustworthy for inference compared to the simpler GLM.

In conclusion, while age has a clear and significant effect on wheezing, the evidence for maternal smoking's impact is not strong and repeatedly inconclusive (across different models). The GLMMs and GEE models are preferred for their ability to handle the data's repeated measures structure and provide more accurate estimates of the effects.